Jingfan Tang

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Education

Shanghai Jiao Tong University Sept 2023 – Aug 2027 (expected)
B.E. in Electrical & Computer Engineering, GPA: 3.83/4.0

Carnegie Mellon University Jan 2026 – May 2026
Department of Electrical & Computer Engineering, College of Engineering

Publications

Goal2Pixel: Grounding Goals to Pixels for Vision-Language Navigation
Muyi Bao, Yuxin Cai, Hang Xu, Zongtai Li, Jinxi He, ***Jingfan Tang***, Chen lv, Ji Zhang, Yaqi Xie, Wenshan Wang
In submission to Conference of Robot Learning (CoRL), 2026

RPV*: Routing with Probabilistic Vertices and Its Use in Autonomous Exploration
Jingfan Tang, Ruizhi Deng, Yunpeng Lyu, Shizhe Zhao, Zhongqiang Ren
International Symposium of Robotics Research (ISRR), 2026

HEHA: Bounded Suboptimal Planning for Constrained Routing and Its Use in Heterogeneous Multi-Robot Exploration of Unknown Environments
Longrui Yan[†], Yiyu Wang[†], ***Jingfan Tang***[†], Yunpeng Lyu, Shizhe Zhao, Chao Cao, Zhongqiang Ren
In submission to IEEE Transactions on Robotics (T-RO), 2026

Research Experience

Undergraduate Researcher, Shanghai Jiao Tong University Mar 2025 – Present

Project: Hierarchical Autonomous Exploration System with Map Prediction

- Developed a modular autonomous exploration framework that integrates map prediction with hierarchical planning, improving exploration efficiency by **7–20%** over state-of-the-art baselines such as TARE and P^2Explore.
- Formulated a **novel uncertainty-aware routing problem** for planning over probabilistic frontiers, and implemented both **optimal and bounded-suboptimal solvers** to mitigate prediction errors and reduce collision risk in partially observed environments.
- Deployed and validated the system on an AgileX Scout Mini robot in cluttered indoor and outdoor environments, demonstrating reliable sim-to-real transfer.

Project: Hierarchical Multi-Robot Exploration System for Heterogeneous Agents

- Built a hierarchical multi-robot exploration framework for task allocation and path planning among heterogeneous agents.
- Improved the local-stage planner, achieving a **16.5%** gain in exploration efficiency and convergence speed over the baseline.
- Designed and conducted simulation experiments to evaluate robustness and scalability in dynamic environments, reducing exploration time by **up to 30%** compared with baselines such as NBVP.

Advisor: Prof. Zhongqiang Ren

Research Intern, Carnegie Mellon University Jan 2026 – Present

Project: Language-Instructed Object Navigation Benchmark

- Developed a benchmark for language-instructed object navigation, integrating landmark-guided instruction following with object-goal navigation to evaluate long-horizon embodied agents.
- Built a trajectory-processing pipeline to extract, filter, and restructure sub-instructions from landmark-based navigation datasets into staged navigation tasks.
- Designed multi-level task difficulty settings based on landmark visibility, semantic guidance, and trajectory structure to assess language grounding and object-search efficiency.
- Implemented rollout and evaluation tools to analyze agent behavior across instruction-following and

object-navigation stages.

Advisor: Prof. Ji Zhang

Skills

- **Languages**: C, C++, Python, Matlab
- **Framework and Operating System**: Linux, Git, ROS (Robot Operating System)
- **Speaks**: Mandarin, English (TOEFL 107)

Honor and Awards

- **The John Wu & Jane Sun Merit Scholarship (Top 2% in overall performance)**, Shanghai Jiao Tong University, 2025
- **Undergraduate Excellent Scholarship (Top 5% in academic performance)**, Shanghai Jiao Tong University, 2024–2025
- **The Yu Liming Scholarship (Top 5% in overall performance)**, Shanghai Jiao Tong University, 2024

Services

- **Teaching Assistant of Course MATH2030J Discrete Mathematics**
  Lecturer: Dr. Runze Cai.
- **Teaching Assistant of Course MATH2550J Honor Multivariate Calculus**
  Lecturer: Dr. Runze Cai.
- **Teaching Assistant of Course ECE2810J Data Structure and Algorithms**
  Lecturer: Dr. Weikang Qian.
- **Organizer of CMU-VLN-Challenge 2026**
- **Reviewer for IROS 2026, Autonomous Robots 2026**
- **Leader and Advisor of GC Student Advising Center**
  Conducted Workshops on graduate school application (2025 Summer), and on freshman course selection (2024 Fall, 2025 Fall).